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MILITARY SPECIFICATION

RESIN, ACRYLIC, DENTAL, FIBERED, DENTURE BASE MATERIAL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers fibered acrylic base resin (polymer) suitable for use in the fabrication of fibered dentures.

1.2 Classification. Fibered denture base polymer covered by this specification shall be of one class (powder and liquid), and of the following types and colors, as specified (see 6.1):

Type I. - Normal impact.
Color A. - Light Pink.
Color B. - Reddish Pink.

Type II. - High impact, Light Pink.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bid or request for proposal form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Personnel Support Center, Directorate of Medical Materiel, DPSC-ATT, 2800 S. 20th St., Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6520

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SPECIFICATIONS

FEDERAL

PPP-B-566	-	Boxes, Folding, Paperboard.
PPP-B-636	-	Boxes, Shipping, Fiberboard.
PPP-B-676	-	Boxes, Setup.
PPP-C-186	-	Container, Packaging and Packing for Drugs, Chemicals, and Pharmaceuticals.

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-	Marking for Shipment and Storage.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bid or request for proposal shall apply.

AMERICAN DENTAL ASSOCIATION (ADA)

Specification No. 12 for Denture Base Polymers.

(Application for copies should be addressed to the Council on Dental Materials and Devices, American Dental Association, 211 East Chicago Avenue, Chicago, IL 60611.)

U. S. DEPARTMENT OF HEALTH, EDUCATION AND WELFARE (HEW), FOOD AND DRUG ADMINISTRATION (FDA)

Federal Food, Drug, and Cosmetic Act and Regulations promulgated thereunder.

(Application for copies should be addressed to the Food and Drug Administration, Washington, DC 20204.)

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3. REQUIREMENTS

3.1 Materials.

3.1.1 Liquid (Types I and II). The liquid shall be methyl methacrylate and ethylene glycol dimethacrylate, with a suitable storage polymerization inhibitor. The liquid shall be clear, colorless, and substantially free from undissolved or particulate matter when examined, without accessory magnification, against a black background and against a white background. Illumination shall be from a light which, at a point 10 inches from its source, provides an intensity of illumination of not less than 100 and not more than 350 footcandles. The liquid shall be a cross-link type monomer, which produces a cross-linked cured resin on polymerization with the powder, when processed in accordance with manufacturer's instructions.

3.1.1.1 Thermal stability of liquid. The liquid shall not discolor or increase in viscosity by more than 10%, when tested as specified in 4.4.8.

3.1.2 Type I powder, Normal impact.

3.1.2.1 Denture base powder. The powder shall be poly (acrylic acid esters), poly (substituted acrylic acid esters), or copolymers or mixtures of the foregoing.

3.1.2.1.1 Color A powder. The denture base powder shall consist of light pink, internally pigmented, spherical particles which shall be readily plasticized to a moldable dough, when properly mixed with the monomer. Cellulose acetate fibers shall be distributed among the particles. The fibers shall be red and violet in color, 7 to 9 denier, 3/8 ± 1/32 inch in length, and each pouch shall contain approximately 4,000 filaments, as determined by a fiber weight of 0.025 to 0.035 grams per pouch.

3.1.2.1.2 Color B powder. The denture base powder shall consist of reddish pink, externally pigmented, spherical particles which shall be readily plasticized to a moldable dough, when properly mixed with the monomer. Cellulose acetate fibers shall be distributed among the particles. The fibers shall be red and violet in color, 7 to 9 denier, 3/8 ± 1/32 inch in length, and each pouch shall contain approximately 10,240 filaments, as determined by a fiber weight of 0.075 to 0.085 grams per pouch.

3.1.2.2 Gingival tone powder. The dispensing container shall be a 15 cubic centimeter, squeeze-type, translucent polyethylene bottle with a screw-type plastic cap containing one half portion of gingival tone, heat-curing polymer. When processed according to instructions furnished by the manufacturer, the polymer shall characterize the gingival anatomy.

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3.1.3 Type II powder, High impact. The powder shall be a graft polymer of polymethyl methacrylate upon butadiene styrene. The denture base powder shall consist of light pink, pigmented particles which shall be readily plasticized to a moldable dough, when properly mixed with the monomer. Polyvinyl chloride filaments shall be distributed among the particles. The filaments shall be red in color, 0.020 to 0.040 millimeters (mm) in diameter, 1/8 ± 1/16 inch in length, and in a concentration of 0.17 ± 0.05 percent by weight.

3.2 Contaminants. The powders shall be free of extraneous material such as dirt or lint that could adversely affect the appearance or properties of the processed resin.

3.3 Properties of monomer-polymer mixture.

3.3.1 Packing readiness. The mixed monomer-polymer shall be ready for packing in 3-1/2 minutes for Type I resin and in 7 minutes for Type II resin, when tested as specified in 4.4.1.1.

3.3.2 Packing plasticity. The mixed monomer-polymer shall intrude into not less than two holes of the die shown in figure 1, to a depth of not less than 0.5 mm, when tested as specified in 4.4.1.2.

3.4 Properties of processed polymer. When mixed as specified in 4.4.1.1, and further processed in accordance with the manufacturer's instructions, the polymer shall yield a satisfactory denture. The processed polymer shall be non-porous (see 3.4.2) and free from surface defects. The processed polymer shall present a smooth surface with a high gloss when polished by the conventional dental methods.

3.4.1 Color. The color of the processed polymer shall conform to color reflectance values as specified in table I, when tested as specified in 4.4.10.

TABLE I. Color reflectance.

Classification	Rd	a	b
Type I, Color A	17.8 - 19.8	17.8 - 19.8	5.5 - 7.0
Type I, Color B	17.5 - 19.5	23.8 - 25.8	6.0 - 8.0
Type II	17.0 - 20.0	17.5 - 19.5	7.0 - 8.5

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3.4.2 Porosity. The specimen plate of figure 2, cut as described in 4.4.5, shall show no bubbles or voids when viewed without magnification.

3.4.3 Sorption. The increase in weight of the polymer shall not be more than 0.8 milligrams per square centimeter (mg/cm^2) of surface, when tested as specified in 4.4.3.

3.4.4 Solubility. The loss of weight of the polymer shall not be more than 0.04 mg/cm^2 of surface, when tested as specified in 4.4.4.

3.4.5 Transverse deflection. The transverse deflection shall comply with the requirements of table II, when tested as specified in 4.4.5.

TABLE II. Transverse deflection.

Load increment		Deflection, mm	
Newtons (N)	Grams	Minimum	Maximum
14.7 - 34.3	1,500 - 3,500	---	2.5
14.7 - 49.0	1,500 - 5,000	2.0	5.5

3.4.6 Color stability. The color change shall be no greater than 2.0 delta E units, when tested as specified in 4.4.6.

3.4.7 Impact strength (Type II only). The minimum impact strength of the Type II polymer shall be 4.8 foot-pounds (ft.-lbs.)/inch, when tested as specified in 4.4.7.

3.5 Toxicity. The manufacturer shall submit a statement that the denture base, produced according to the instructions accompanying the material, has no toxic effect on the normal, healthy person.

3.6 Shelf life. The shelf life of the resin shall be a minimum of 24 months from date of manufacture when stored at 59° to 86°F. Not less than 20 months shall remain at the time of delivery to the government. The inspection test date shall be 24 months from date of manufacture.

3.7 Manufacturer's instructions. Adequate and accurate instructions for storing, preparing, and processing the resin shall be provided. These instructions shall accompany each package and shall be such that denture base molding can be carried out with the equipment generally available to the dental laboratory of the general practitioner. These instructions shall include as appli-

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cable: recommended condition of storage where shelf life is limited, powder-liquid ratio, the time and procedure required to mix the material properly, time required for the material to reach the proper packing plasticity at $23^{o} \pm 1^{o}C$, working time during which packing may proceed, the material used to prepare the mold (for example, gypsum or hydrocolloid), temperature of flask when the polymer is packed, the proper curing temperature, time to bring the flask to recommended curing temperature(s), time to hold at curing temperature, method of cooling flask, and time at which the denture may be deflasked. The manufacturer's instructions shall conform to those specified in 4.4.2. In addition to the instructions given in 4.4.2, the manufacturer may provide alternate techniques. The manufacturer's instructions shall include a method for repairing the processed material. The instructions shall also include a statement of caution against prolonged handling of the uncured mixture with bare hands and against inhalation of monomer vapor.

3.8 Put-up.

3.8.1 Type I. The Type I resin shall be furnished as a package consisting of one light-resistant glass bottle of liquid monomer containing not less than 120 milliliters; 10 pouches of polymer powder containing not less than 21 grams in each pouch; one dispensing container of gingival tone polymer containing not less than 7 grams; and the manufacturer's instructions.

3.8.2 Type II. The Type II resin shall be furnished as a package consisting of one light-resistant glass bottle of liquid monomer containing not less than 120 milliliters; one plastic bottle containing not less than 231 grams of powder; a minimum of 15 heat-sealable plastic bags for dentures; a powder measure; a liquid graduate; and the manufacturer's instructions.

3.9 Certification. Suppliers shall furnish, at time of award, a dated certificate and data showing that the denture base resin they propose to furnish is the same in quality and brand as that appearing on the list of certified materials for American Dental Association Specification No. 12 for denture base polymers. Compliance with this certification requirement does not absolve the contractor from complete compliance with the requirements of this specification.

3.10 Workmanship. Workmanship shall be first class throughout. The products shall be free from defects which detract from their appearance or impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

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4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the fibers and packaging materials, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the name of the material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, the specification requirements, the NSN, and the Item Identification.

4.2 Sampling.

4.2.1 Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105. The inspection levels and acceptable quality levels (AQL's) shall be as indicated in table III. The unit of product for examination shall be one unit of issue.

TABLE III. Examination criteria.

Type defect	Inspection level	AQL (percent defective)
Major	S-2	1.0
Minor	S-2	2.5

4.2.2 Sampling for tests. Sampling for tests shall be conducted on each lot or control number in accordance with table IV. If any test determination for a lot does not comply with the requirements specified, the lot shall be rejected.

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TABLE IV. Test criteria.

Characteristic	Requirement	Test procedure	Minimum number of determinations per lot or batch number
Thermal stability	3.1.1.1	4.4.8	1
Packing readiness	3.3.1	4.4.1.1	2
Packing plasticity	3.3.2	4.4.1.2	2
Glossy surface	3.4	1/	2
Polish	3.4	1/	2
Color	3.4.1	4.4.10	2
Porosity	3.4.2	4.4.5	1 2/
Sorption	3.4.3	4.4.3	1 3/
Solubility	3.4.4	4.4.4	1 3/
Transverse deflection	3.4.5	4.4.5	1 4/
Color stability	3.4.6	4.4.6	1
Impact strength	3.4.7	4.4.7	1
Weight of fibers	3.1.2.1, 3.1.3	4.4.11	2
Leakage test	5.1.2.1, PPP-C-186	4.4.9, PPP-C-186	See 4.2.2.1
Net content of powder and liquid	3.8, 5.1.1	Standard measuring instruments	1
Compatibility of container	5.1.2.1	4.4.12	2

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1/ The polymer processed according to instructions furnished by the manufacturer (3.7).

2/ Each determination shall be the average of 5 specimens.

3/ Each determination shall be the average of 2 specimens.

4/ Each determination shall be the average of at least 3 of the 5 specimens.

4.2.2.1 Sampling for leakage test. Sampling for leakage test shall be in accordance with MIL-STD-105, inspection level S-2. The acceptance number shall be zero.

4.3 Examination. The resin shall be examined to determine compliance with requirements contained in this specification.

4.3.1 Classification of defects. The resin shall be examined for defects including, but not limited to, those listed in table V.

TABLE V. Classification of defects.

Categories	Defects
Major	
101	Type not as specified.
102	Color not as specified.
103	Material not free from foreign matter.
104	Broken or cracked bottle.
105	Caps loose on bottles.
Minor	
201	Manufacturer's instructions not present in unit, incorrect, incomplete, or not applicable to material furnished.
202	Put-up not as specified.

4.3.2 Packaging inspection. The inspection of the unit package, packing, and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements. All equipment shall be conditioned and all tests shall be conducted at $23^{o} \pm 2^{o}C$ unless otherwise specified.

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4.4.1 Mixing and packing characteristics.

4.4.1.1 Mixing test. The polymers shall be mixed as directed in the instructions accompanying the package, except that the mixing temperature shall be 23°± 2°C. Unless otherwise specified in the instructions, the polymer shall be considered ready for packing when the plasticity is such that the polymer separates cleanly from the walls of the glass mixing jar.

4.4.1.2 Packing test. The perforated brass die (figure 1) and glass plate, approximately 5 mm thick and 60 mm square, shall be conditioned to the packing temperature specified in the manufacturer's instructions (3.7) and held within ± 2.5°C until the intrusion measurement is begun. The packing test shall be started 60 minutes (for Type I) or 5 minutes (for Type II) after the material reaches proper plasticity as specified in 4.4.1.1. An 8 to 10 gram sample of resin shall be prepared in accordance with the manufacturer's instructions. At the recommended initial packing time, it shall be shaped to a thickness of approximately 5 mm and shall be placed on the die conditioned to the specified packing temperature in the environment prescribed by the manufacturer. The resin shall be covered with a sheet of polyethylene and immediately the glass plate and a 5-kg mass shall be placed on top. Ten minutes later the load shall be removed and the depth of intrusion into each hole shall be measured to a precision of 0.1 mm before the material is stripped from the die.

4.4.2 Preparation of test specimens of cured polymer for sorption, solubility, and transverse deflection tests. The flask shall be immersed in boiling water and then successively flushed with a biodegradable detergent solution and clear boiling water. After the mold has cooled until it can be held comfortably in the bare hand, it shall be painted with one or more coatings of undiluted alginate compound tinfoil substitute. Preparation of molds for resins requiring warm or heated molds shall be conducted according to the manufacturer's instructions. Ten minutes after the proper plasticity as specified in 4.4.1.1 has been reached, the flasked specimen (figures 2 and 3), held in a flask clamp, shall be submerged in water at 73°± 1°C for 1-1/2 hours and then immersed for 30 minutes in boiling water. When the heating schedule has been completed, the flask in the clamp shall be cooled in air at 23°± 10°C for 30 minutes or more, and then immersed in water at 23°± 10°C for 15 minutes.

4.4.3 Water sorption test. The water sorption test shall be made in duplicate on disks 50 ± 1 mm in diameter and 0.5 ± 0.05 mm thick. These specimens shall be formed in a stainless steel-gypsum mold (figure 3). The gypsum cover surface for the mold shall be formed against a flat glass or metal surface to obtain a smooth, plain surface. The resin shall be introduced into the mold as prescribed by the manufacturer, except that there shall be no trial packing which will disturb the tinfoil substitute film. The disk thus obtained will be approximately 0.88 mm thick. Each specimen shall then be reduced to tolerance

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by removing at least 0.15 mm from each surface by accepted metallographic hand or machine grinding procedures employing abrasive papers in successive grits of 120, 240, 400, and 600. An assembly for reducing the specimen thickness is shown in figure 4. The abrasive papers shall be flooded with water during the grinding procedures. All surfaces of the disk shall be smooth, and the top and bottom shall be flat. The disks shall be dried in a desiccator containing thoroughly dry anhydrous calcium sulfate ($CaSO_4$) or silica gel (freshly dried at 130°C) at 37° ± 2°C for 24 hours, removed to a similar desiccator at room temperature for one hour, and then weighed with a precision of 0.2 mg. This cycle shall be repeated until the weight loss of each disk is not more than 0.5 mg in any 24-hour period. The disks shall then be immersed in distilled water at 37° ± 1°C for seven days, after which time the disks shall be removed from the water with tweezers, wiped with a clean, dry hand towel until free from visible moisture, waved in the air for 15 seconds, and weighed one minute after removal from the water. The value for water sorption shall be calculated as follows for each disk: [mass after immersion (mg) - conditioned mass (mg)] / surface area (cm^2) = sorption (mg/cm^2). The average of the determined values for the two disks shall be recorded to the nearest 0.01 mg/cm^2. The final value (average of two determinations) shall be rounded to the nearest 0.1 mg/cm^2. If the final value falls midway between two numbers, the even number shall be recorded.

4.4.4 Solubility test. After the final weighing (4.4.3), the disks shall be reconditioned to constant weight in the desiccator at 37° ± 2°C as was done in 4.4.3. The value for solubility shall be calculated as follows for each disk: [conditioned mass (mg) (4.4.3) - reconditioned mass (mg) (4.4.4)] / surface area (cm^2) = solubility (mg/cm^2). The final value (average of two determinations) shall be rounded to the nearest 0.01 mg/cm^2. If the final value falls midway between two numbers, the even number shall be recorded.

4.4.5 Transverse deflection test. A specimen plate of polymer shall be processed in a gypsum mold as described in 4.4.2. Specimens for the transverse test shall be machined from this plate to the dimensions shown in figure 5. The plate (figure 2) shall be sawed lengthwise into five equal strips with a band saw operated at a speed of approximately 100 M-min (328 feet/min). The strips may then be sanded slightly on the molded surfaces, if necessary, to remove imperfections. All subsequent strips shall be machined on both the molded surfaces and the edges so that the dimensions are slightly oversize. Care shall be taken to avoid overheating the specimen. One preferred technique is to use a 50-mm diameter fly cutter, operating at a spindle speed of approximately 1,800 rpm and at a table feed of about 15 cm/min. As an alternative, a slab mill or an end mill may be used. Rough cuts shall be approximately 0.08 mm deep, and finish cuts shall be 0.03 mm deep. The strips shall be mounted side by side in a vise and the edges machined parallel in order to remove all band saw marks.

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The strips at this point shall be oversize in width. One or more strips shall be mounted in a vise and the faces milled to the required thickness (2.50 ± 0.03 mm). All five strips shall then be clamped face to face in a vise and the edges milled to the required width (10.00 ± 0.03 mm). The specimens shall be stored in distilled water at 37°± 1°C for 50 ± 2 hours immediately before testing. The specimen shall be mounted in an appropriate calibrated instrument capable of a uniform loading rate and equipped with a device for measuring deflection within 0.01 mm at the center of the specimen, such as shown in figure 5. The loading nose and the two supports shall have highly polished cylindrical surfaces 3.2 mm in diameter and shall be parallel to within 0.1 mm over a 10-mm length. The distance between the centers of the supports shall be 50 ± 0.025 mm.

TABLE VI. Schedule of loading and reading.

Time		Load		Dial reading 3/
Minutes	Seconds	Newtons (N)	Grams	mm
0	00	14.71	1,500	
0	30	14.71	1,500	A
1	00	19.61	2,000	
1	30	19.61	2,000	
2	00	24.52	2,500	
2	30	24.52	2,500	
3	00	29.42	3,000	
3	30	29.42	3,000	
4	00	34.32	3,500	
4	30	34.32 1/	3,500	B
5	00	39.22	4,000	
5	30	39.22	4,000	
6	00	44.13	4,500	
6	30	44.13	4,500	
7	00	49.03	5,000	
7	30	49.03 2/	5,000	C

1/ Deflection at 34.32N=B-A.
2/ Deflection at 49.03N=C-A.
3/ Readings shall be made to the nearest 0.01 mm.

The loading nose shall be midway between the supports to within 0.025 mm. Means should be provided to prevent misalignment of the specimen on the supports. The specimen shall be immersed in water during testing. An initial load, consisting of the free-moving parts of the apparatus (figure 5), plus sufficient lead shot to make a load of 14.71 N, shall be placed on the specimen. Each 4.90-N increment of shot shall be added at a uniform rate during the last 30 seconds of each minute.

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The schedule shown in table VI shall be used in applying the loads and in making the observations. The differences between the deflections at the initial load and at specified loads shall be taken as the deflections of the specimen. The recorded deflections shall be reported to the nearest 0.1 mm. The value for transverse deflection shall be reported as the average of three or more specimens from a lot of five. If the determined value falls midway between two successive numbers, the even number shall be recorded.

4.4.6 Color stability test. The color stability test shall be made on a disk 50 + 1 mm in diameter and 3.00 + 0.25 mm thick. The disk, prepared as specified in 4.4.10, shall be exposed to the radiation of a lamp (figure 6) with an S-1 bulb which has been in use not less than 50 nor more than 400 hours. This light source (S-1 bulb), rated at 400 watts, shall be a combination tungsten-filament mercury arc enclosed in glass which filters out ultraviolet light below 280 nm. The specimen shall be carried on an aluminum disk resting on a turntable operating at 33 rpm, and shall be centered under the S-1 bulb. The specimen shall be held by screws approximately 5 mm above the disk and 12.7 cm from the center, and shall be maintained between 60° and 65°C. The plane of the top surface of the specimen shall be 17.8 cm from the bottom of the S-1 bulb. After exposure for 24 hours, the exposed specimen, compared with the unexposed specimen, shall not show more than a delta E of 2.0 units (see 4.4.10).

4.4.7 Impact strength test (Type II only). Impact strength shall be determined on specimens prepared in a 2-1/2 inch by 2-1/2 inch by 0.2 inch brass mold lined with aluminum foil. The powder and liquid shall be mixed in accordance with the manufacturer's instructions and the mold filled with the uncured mix. The filled mold shall then be pressure seated in a flask, clamped and cured in accordance with the manufacturer's instructions. The cured resin shall be removed from the mold and cut into five pieces with a band saw. The surfaces of the specimens shall be sanded to remove flash. The specimens shall then be reduced to 2.65 $\pm$ 0.05 mm in thickness and 10.15 $\pm$ 0.05 mm in width on a milling machine with a 50 mm diameter fly cutter operating at 1800 rpm, and a table feed of about 15 centimeters per minute. The specimens shall then be finished to dimensions of 2.50 $\pm$ 0.03 mm in thickness and 10.00 $\pm$ 0.03 mm in width by use of a fine abrasive paper on a hard flat surface. The specimens shall then be stored in water at 37° $\pm$ 2°C for 48 hours. The five specimens shall be removed and each cut in half to a length of 1-1/8 inch with a band saw to form 10 specimens. The specimens shall be clamped securely to a depth of 1.7 mm in the vise of a model TM 52004 impact tester, manufactured by Testing Machine, Inc., or equivalent tester. The edge of the specimen shall face the swing of the pendulum. The average of the results on the 10 specimens shall be converted to a one inch width basis and reported as one determination of impact strength.

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4.4.8 Thermal stability test. The color and viscosity change of the liquid shall be determined on the liquid heated in the dark in a closed container at 60° ± 2°C for 48 hours. The color change shall be determined by visual comparison of the liquid before and after heating by the use of 100 ml Nessler color comparison tubes. The change in viscosity shall be determined at 25°C in a viscometer such as the number 25 type G Ostwald-Cannon-Fenke viscometer. The viscometer and samples shall be preconditioned for 2 hours in a 25°C constant temperature bath. A determination shall be the average of the results of three specimens.

4.4.9 Leakage test. The bottle of liquid, with closures and seals in place, shall be subjected to the leakage test, as specified in PPP-C-186 for immediate containers holding flammable liquids, except that sampling for this test shall be as specified in 4.2.2.1.

4.4.10 Color determination test. The color of the cured resin shall be determined on a disk, 50 ± 1 mm in diameter and 3.00 ± 0.25 mm in thickness, which has been in a mold lined with aluminum foil. The powder and liquid shall be mixed in accordance with the manufacturer's instructions and the mold filled with the uncured mix. The filled mold shall then be pressure seated in a flask, clamped and cured in accordance with the manufacturer's instructions. The cured resin shall be removed from the mold. The foil shall be removed from the resin and any flashing shall be sanded off. The color reflectance determination shall be made on the specimen using a Gardner Color Difference Meter Model AC-1 with a one inch diameter aperture.

4.4.11 Weight of fibers test. The contents of one pouch of Type I polymer (21 grams), or one bottle of Type II polymer, as applicable, shall be placed on a United States Standard Sieve No. 25 and shall be gently vibrated to separate the fibers from the polymer. This procedure shall be repeated on the sample until there is an effective separation of fibers. The collected fibers shall then be weighed.

4.4.12 Compatibility of container test. The inner seal shall be removed. The container shall be allowed to stand capped for 24 hours, after which it shall be examined for evidence of incompatibility.

5. PACKAGING

5.1 Packaging. Packaging shall be level A or C, as specified (see 6.1).

5.1.1 Unit of issue.

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5.1.1.1 Type I. One package consisting of one bottle of liquid monomer containing not less than 120 milliliters; 10 pouches of polymer powder containing not less than 21 grams in each pouch; one dispensing container of light colored polymer containing not less than 7 grams; and instructions, as specified, constitutes one unit of issue.

5.1.1.2 Type II. One package containing one bottle of liquid (120 milliliters), one bottle of powder (231 grams), a minimum of 15 heat-sealable plastic bags, a powder measure, a liquid graduate, and instructions, as specified, constitutes one unit of issue.

5.1.2 Level A.

5.1.2.1 Immediate containers. Liquids shall be packaged in narrow-mouth, light-resistant glass bottles of a suitable size and design. Type II powder shall be packaged in wide-mouth plastic bottles of a suitable size and design. Type I powder shall be packaged in sealed poly-laminated pouches of a suitable size, design, and weight. Bottles shall be closed by means of a continuous threaded plastic or black-phenolic cap. Caps shall have a tight-fitting, impervious, resilient liner which must remain in the cap when the cap is removed from the bottle. The liner shall be aluminum faced, or polyethylene, or a high density cone-shaped liner. Bottles shall be further sealed with an inner seal, which shall cover and adhere to the mouth of the bottle and be capable of preventing loss of contents, should the cap become loosened. An inner seal shall not be required if a cone-shaped liner is used. The container closure and liner shall not interact physically or chemically with the substance contained, such as to be altered by or to alter the strength, quality, or purity of the contents.

5.1.2.2 Unit package. Each unit of issue shall be packaged in a box conforming to PPP-B-566, PPP-B-636, class domestic, or PPP-B-676. Bottles shall be protected with flexible, corrugated fiberboard sleeves or with a suitable interior fitting that will secure the bottles in a vertical position and protect the bottles against damage. Closure shall be as specified in the appendix of the applicable box specification.

5.1.3 Level C. Each unit of issue shall be packaged in a manner that will afford adequate protection against deterioration and physical damage during shipment from the supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.1).

5.2.1 Level A. Twelve unit packages, containing material of one type and color only, shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-636, class weather-resistant. Closure and strapping shall be as specified in the appendix of the box specification.

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5.2.2 Level B. Twelve unit packages, containing material of one type and color only, shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-636, class domestic. Closure shall be as specified in the appendix of the box specification.

5.2.3 Level C. The packaged resin material shall be packed in shipping containers in a manner that will afford adequate protection against damage during direct shipment from the supply source to the first receiving activity. These packs shall conform to the applicable carrier's rules and regulations.

5.3 Marking.

5.3.1 Immediate containers. Immediate containers shall be marked in accordance with the contractor's commercial practice and the requirements of the Federal Food, Drug, and Cosmetic Act. In addition, marking shall include the lot number, and on the bottles of liquid shall also include the legend "FLAMMABLE" and "STORE BETWEEN 16° - 27°C (60° - 80°F)".

5.3.2 Unit package and exterior containers. Each unit package and each exterior container shall be marked as specified in MIL-STD-129. The lot number shall be shown. Type II shelf life marking as specified in MIL-STD-129 shall also be shown. Marking shall also include the legend:

"KEEP AWAY FROM HEAT, SPARKS AND OPEN FLAME"

AND

"STORE BETWEEN 16° - 27°C (60° - 80°F)".

6. NOTES

6.1 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

a. Title, number, and date of this specification.
b. Type and color of resin required (see 1.2).
c. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 This specification covers the following items appearing in the Federal Supply Catalog:

NSN	Item Identification	Specification Designation
6520-00-420-5750	RESIN, ACRYLIC, DENTAL, Fibered, Denture Base Material, Reddish Pink, 210 Gram.	Type I, Color B

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NSN	Item Identification	Specification Designation
6520-00-935-4014	RESIN, ACRYLIC, DENTAL, Fibered Denture Base Material, Light Pink, 210 Gram.	Type I, Color A
6520-01-025-5021	RESIN, ACRYLIC, DENTAL, Denture Base Material, Pink, High Impact, 231 Gram.	Type II

6.3 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Review activity:
DoD-MB

Preparing activity:
DLA-DM

Project No. 6520-1477

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BRASS DIE FOR PACKING TEST

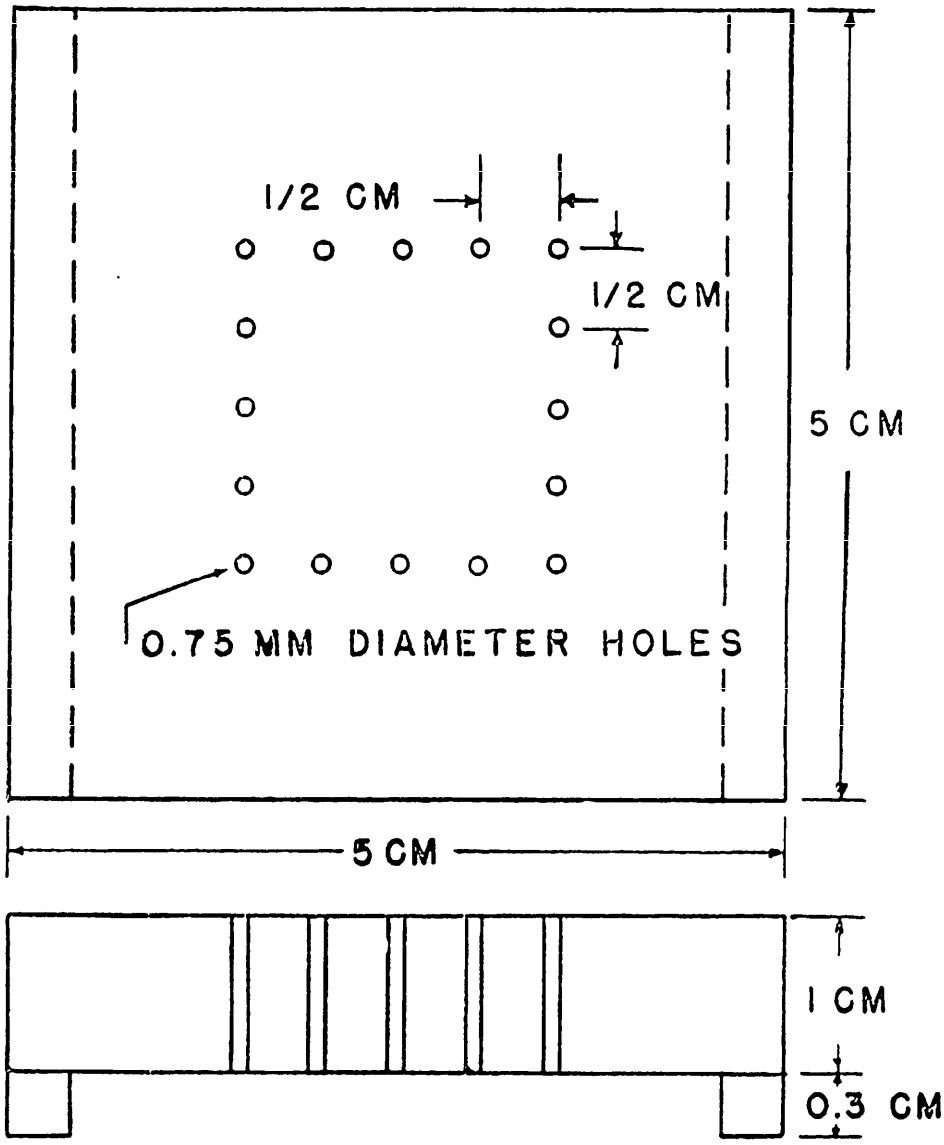
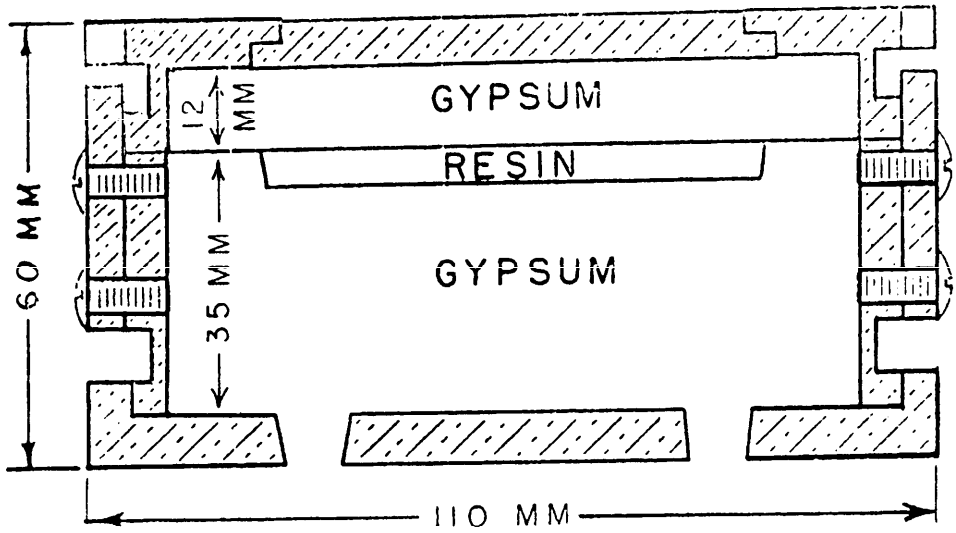
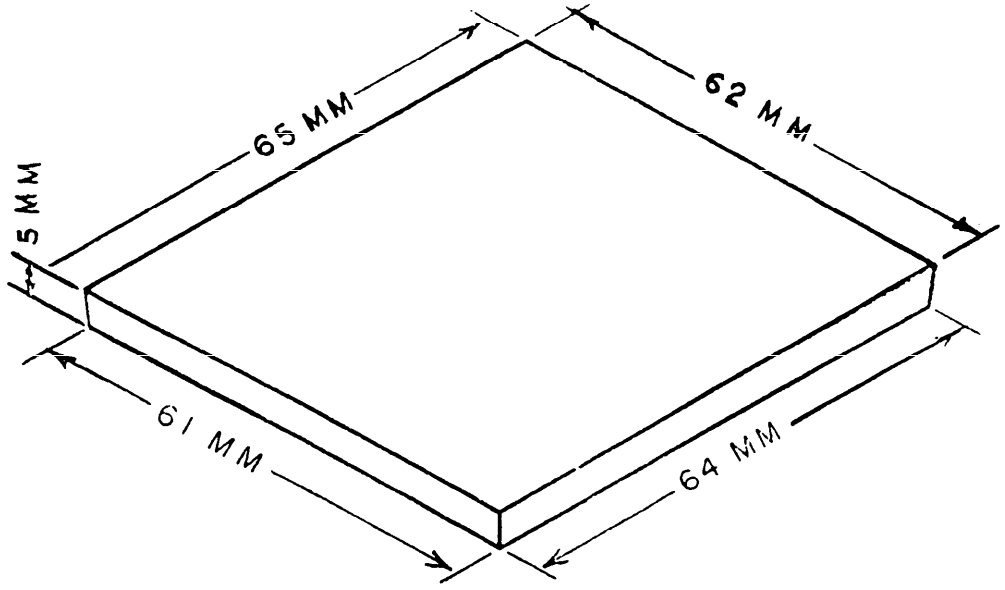


Figure 1

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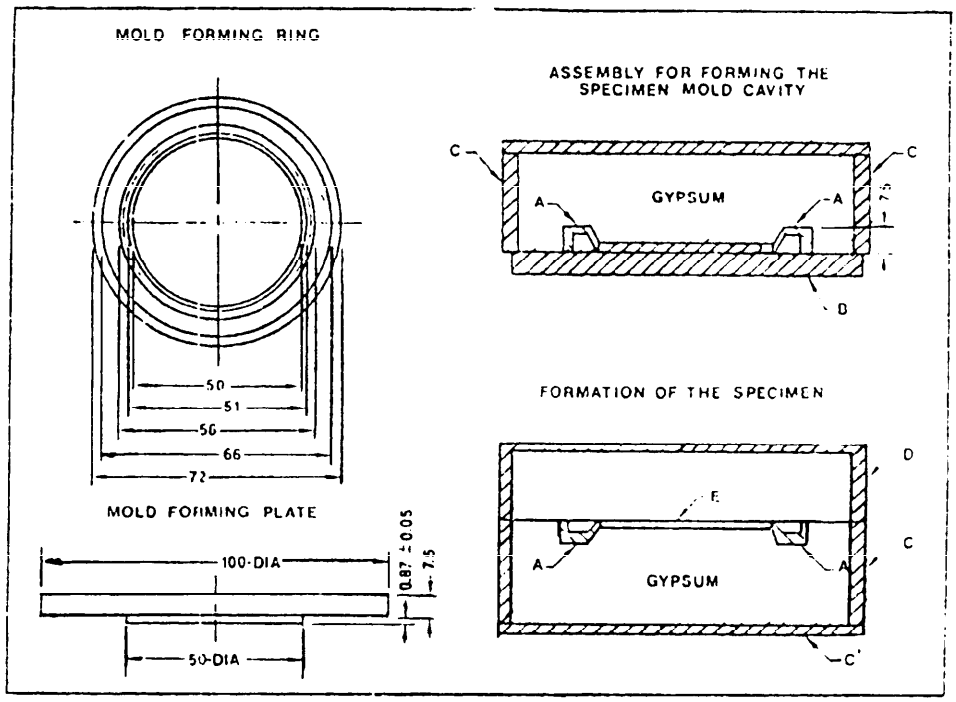
DENTURE FLASK



SPECIMEN PLATE

Figure 2

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SYSTEM FOR FORMING SORPTION-SOLUBILITY DISKS FROM HEAT-PROCESSED RESINS.

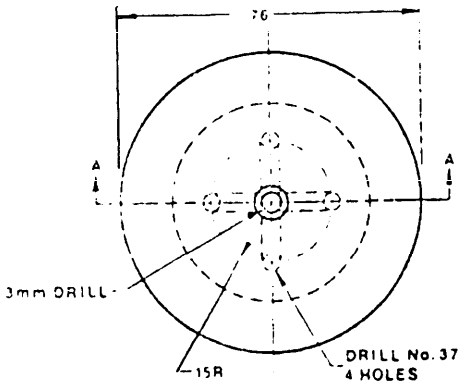
A. mold forming ring;
B. mold forming plate;
C. upper half of denture flask;
D. lower half of denture flask;
E. specimen.

Mold forming ring and plate of stainless steel. All dimensions in millimeters.

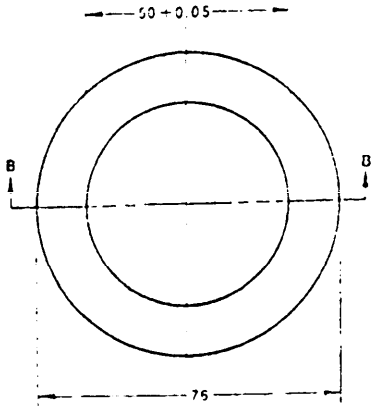
Figure 3

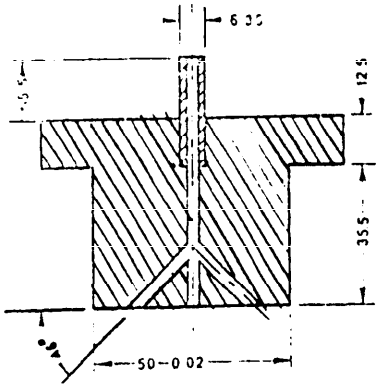
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SPECIMEN RETAINER (Brass)

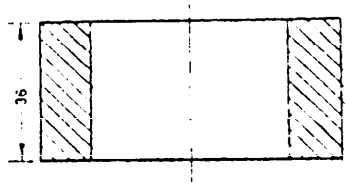


RETAINER GUIDE (Stainless steel)



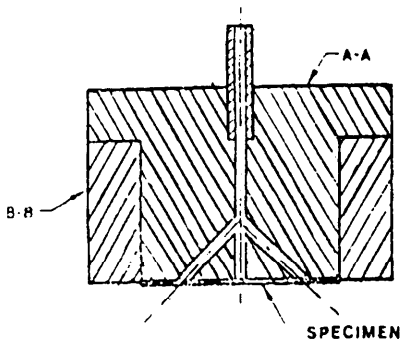


SECTION A-A



SECTION B-B

All Dimensions in mm.



ASSEMBLY FOR REDUCING THICKNESS OF SPECIMEN

ASSEMBLY FOR HOLDING SPECIMEN DURING REDUCTION FOR THICKNESS.

Figure 4

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TRANSVERSE TESTING MACHINE

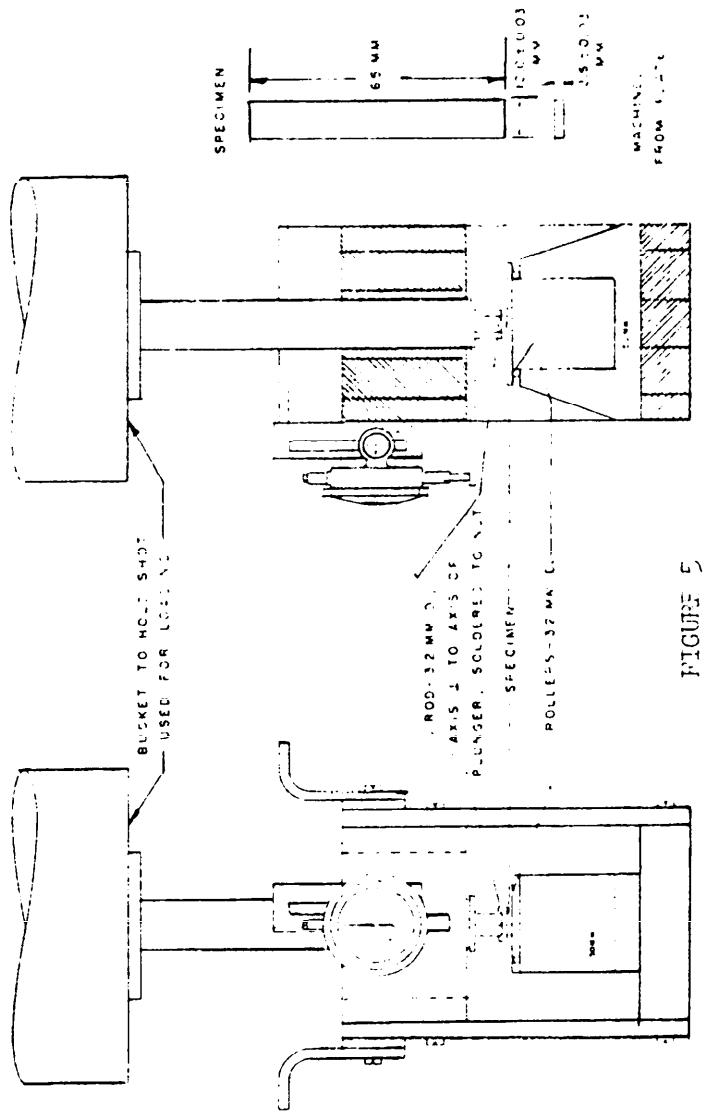
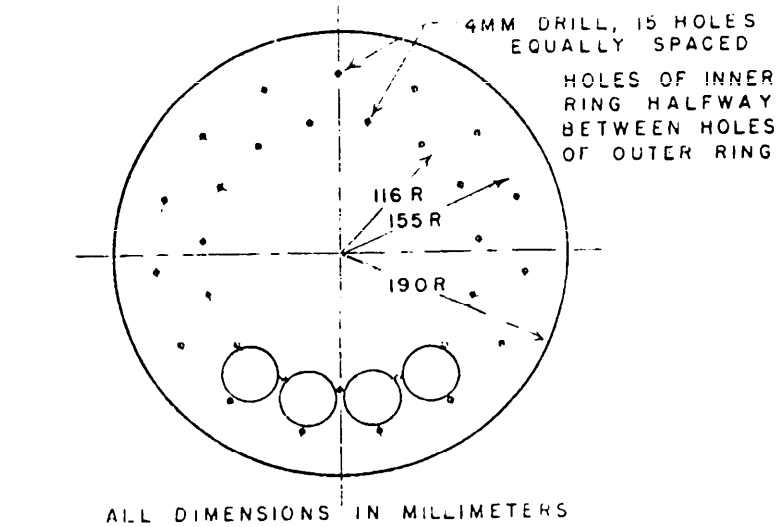


FIGURE 5

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LAMP TO TEST FOR COLOR STABILITY



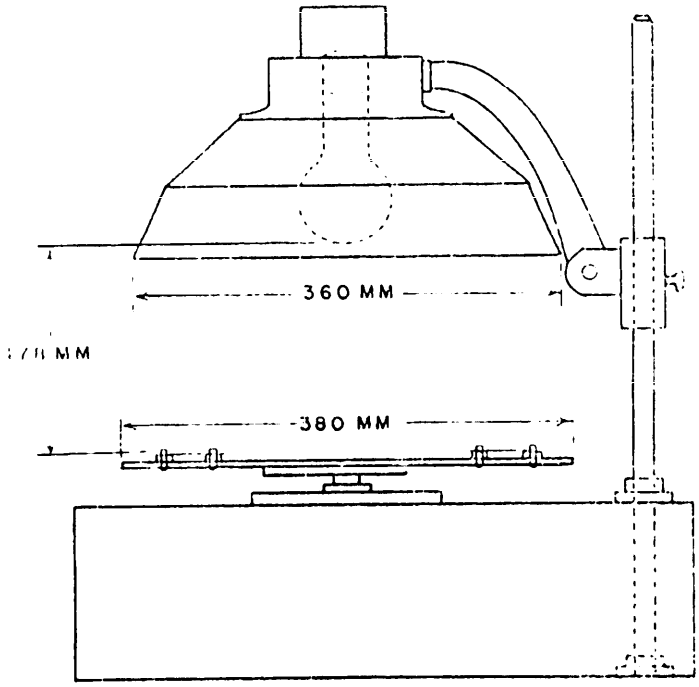


Figure 6

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS. The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers' vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE
MIL-R-37770A RESIN, ACRYLIC, DENTAL, FIBERED, DENTURE BASE MATERIAL

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*	TELEPHONE NO.
	DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

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